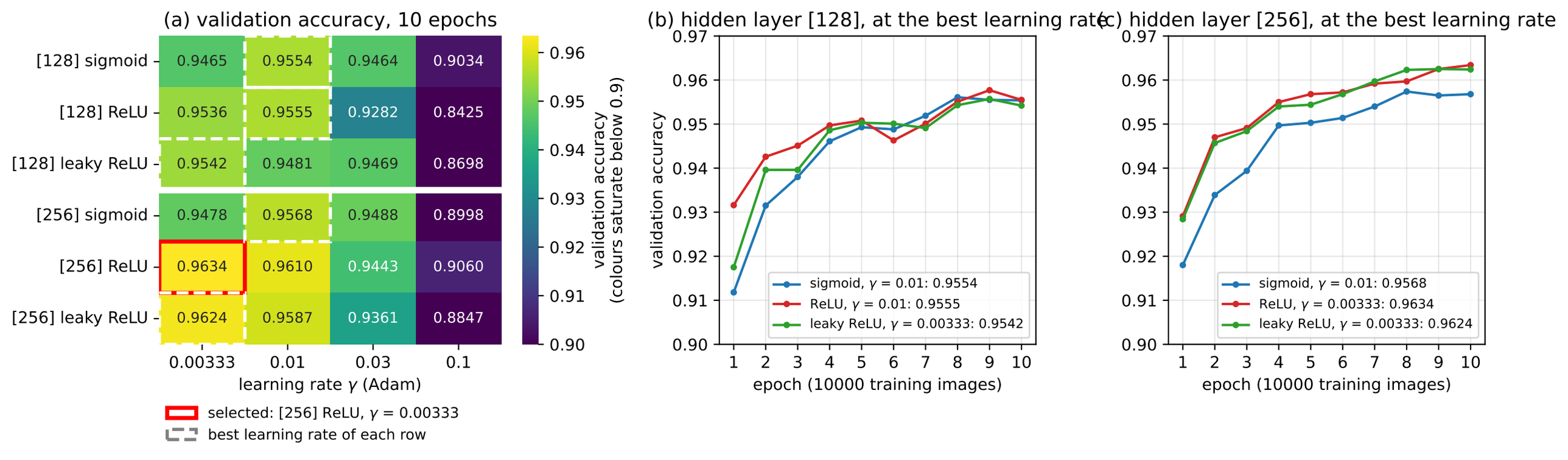
(a) validation accuracy, 10 epochs
[128] sigmoid
[128] ReLU
[128] leaky ReLU
[256] sigmoid
[256] ReLU
[256] leaky ReLU
0.9465
0.9554
0.9464
0.9034
0.9536
0.9555
0.9282
0.8425
0.9542
0.9481
0.9469
0.8698
0.9478
0.9568
0.9488
0.8998
0.9634
0.9610
0.9443
0.9060
0.9624
0.9587
0.9361
0.8847
0.00333
0.01
0.03
0.1
learning rate γ (Adam)
validation accuracy
(colours saturate below 0.9)
0.96
0.95
0.94
0.93
0.92
0.91
0.90
selected: [256] ReLU, γ = 0.00333
best learning rate of each row
(b) hidden layer [128], at the best learning rate
validation accuracy
epoch (10000 training images)
sigmoid, γ = 0.01: 0.9554
ReLU, γ = 0.01: 0.9555
leaky ReLU, γ = 0.00333: 0.9542
(c) hidden layer [256], at the best learning rate
epoch (10000 training images)
sigmoid, γ = 0.01: 0.9568
ReLU, γ = 0.00333: 0.9634
leaky ReLU, γ = 0.00333: 0.9624